

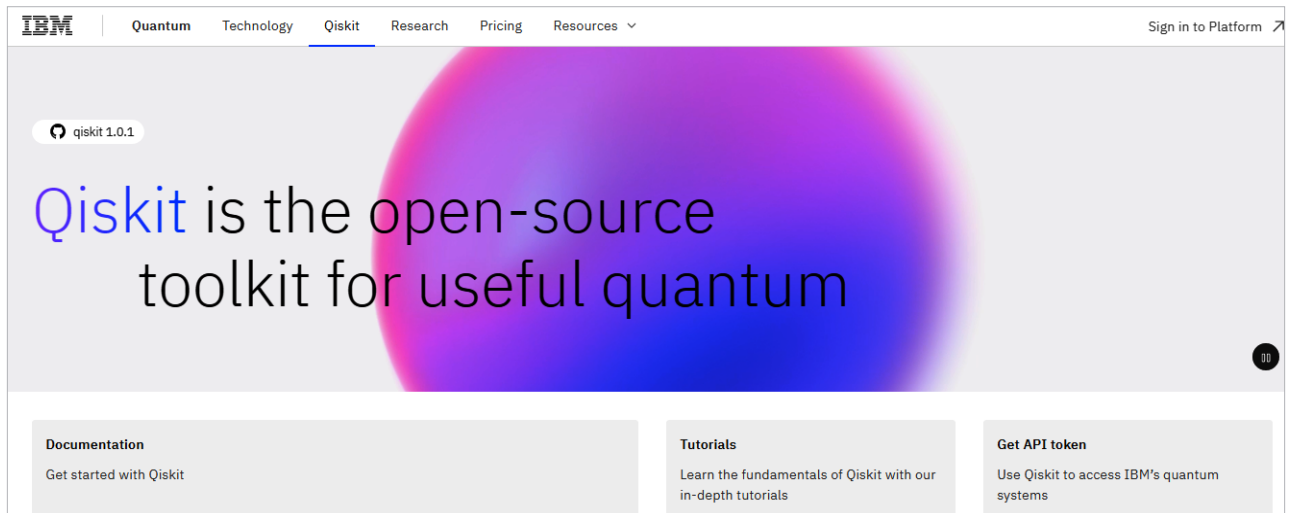


Figure 2: IBM Qiskit for quantum computing

Microsoft's Quantum Development Kit includes Q (Q Sharp), a domain-specific programming language for quantum computing. Developers can use Q to write quantum algorithms and test them using simulators provided in the kit. It also integrates with Visual Studio and offers extensive documentation and learning resources. The kit includes a quantum simulator for testing and debugging quantum algorithms locally before running them on quantum hardware. Microsoft also offers extensive documentation, tutorials, and learning resources to support developers in exploring quantum computing.

Google Cirq

<https://quantumai.google/cirq>

Cirq is an open source framework developed by Google for writing, simulating, and running quantum circuits. It is designed to be flexible and suitable for both researchers and developers. Cirq supports Python and provides tools for quantum algorithm development and hardware interaction. It offers tools for creating and manipulating quantum circuits, as well as interfaces for interacting with quantum hardware. The framework is designed to be flexible and extensible, making it suitable for both algorithm development and hardware testing. Google offers comprehensive documentation, tutorials, and examples to help users get started with Cirq.

Rigetti Forest

<https://www.rigetti.com/forest>

Rigetti Forest is a full-stack quantum computing platform offered by Rigetti Computing. It includes tools for writing quantum programs in Quil (Quantum Instruction Language) and executing them on Rigetti's quantum processors or simulators. The platform supports Quil, Rigetti's quantum instruction language, which allows users to define quantum circuits and execute them on Rigetti's quantum processors or simulators. Rigetti Forest also provides access to Quantum Cloud Services, enabling users to scale up their quantum computations and run them remotely on Rigetti's cloud-based infrastructure.

Amazon Braket

<https://aws.amazon.com/braket>

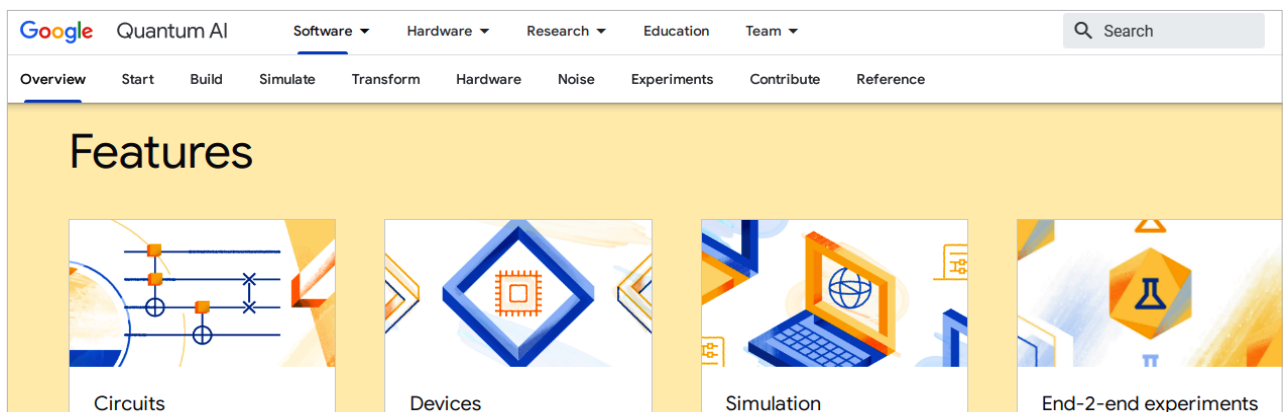


Figure 3: Google quantum AI platform